UNIT – II : Geometric Modeling

1. Introduction

- **Geometric modeling** = mathematical description of the shape (geometry + topology) of an object so that a computer can store, display, analyse and manufacture it.
- The model is the heart of CAD: the same model feeds drafting, analysis (FEA), NC tool-path generation, mass-property calculation and rendering.
- Three types of CAD models:
  - **Wire-frame model** – edges / curves only (skeleton).
  - **Surface model** – skin of the object defined by surface patches.
  - **Solid model** – complete volume with inside / outside information.
- Uses of geometric modeling: design, mass properties, interference check, finite-element mesh, machining simulation, animation.

2. Representation Techniques

- **Explicit (non-parametric)**: $y = f(x)$, $z = g(x)$ – one variable gives the others.
  - Example: straight line $y = mx + c$; circle $y = \pm\sqrt{(r^2 - x^2)}$.
- **Implicit (non-parametric)**: $f(x, y) = 0$ (or $f(x, y, z) = 0$).
  - Example: circle $x^2 + y^2 - r^2 = 0$; sphere $x^2 + y^2 + z^2 = r^2$.
- **Parametric**: every coordinate is a function of a parameter(s): $x = x(u)$, $y = y(u)$, $z = z(u)$ with u from 0 to 1 (curve) or $P(u, v)$ (surface).
  - Circle: $x = r \cos u$, $y = r \sin u$; line: $P(u) = P1 + u(P2 - P1)$.

Parametric vs non-parametric

Non-parametric	Parametric
Depends on the axes; slope may be infinite (vertical line)	Independent of axes; tangent vector always finite
Multi-valued curves (circle) need several equations	Closed / multi-valued curves easy, one equation set
Bounded segments need extra limits	Bounds are built in ($0 \le u \le 1$)
Point on curve found by solving equation	Points generated directly by varying u
Difficult transformations	Simple transformations of control points

CAD systems prefer the **parametric** form: easy to compute, plot, transform, and to join curves.

3. Wire-frame Modeling

- Object represented by **points, lines, circles, arcs and curves** (edges); no surface or volume information.
- Types: 2D, 2½D (extruded) and 3D wire frames.
- Data stored: a **vertex table** (coordinates) and an **edge table** (pair of vertices).

Advantages

- Simple, needs little memory and computing time; fast to draw and edit.
- Gives all views easily; good for rough design, layouts and as the base for surface / solid models.

Limitations

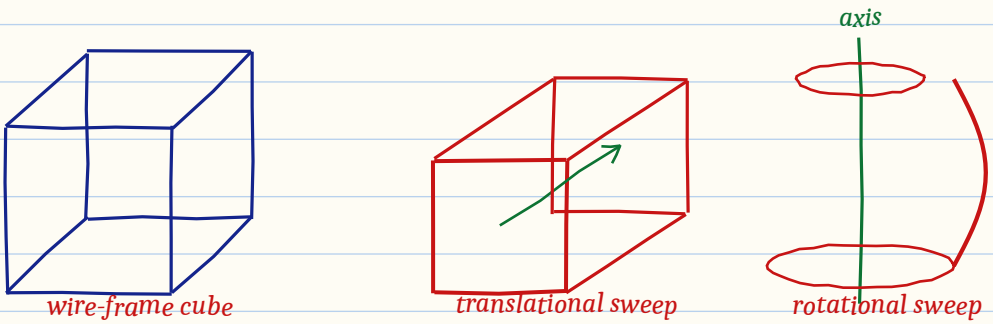
- **Ambiguity** – the same wire-frame can be interpreted as several solids (e.g. Necker cube).
- No hidden-line removal automatically; cannot calculate volume / mass; no section views, no shading; no information for NC machining of faces; interference checking not possible.

4. Construction Methods (creating wire-frame / basic geometry)

- **Points**: by coordinates, by intersection of lines, at the end / mid-point of an entity, on a curve at distance.
- **Lines**: two points; point + angle + length; parallel / perpendicular / tangent to another entity; horizontal / vertical.
- **Circles & arcs**: centre + radius; three points; two points + radius; tangent to two

entities (TTR); fillet / chamfer.

- **Ellipse, parabola, hyperbola** (conics) by centre / foci and axes.
- Editing: trim, extend, offset, mirror, copy, array, scale, rotate.
- **Sweeping**: translational sweep (extrude a profile along a line) and rotational sweep (revolve around an axis).



5. Synthetic Curves (free-form curves)

- Analytic curves (line, circle, conic) cannot describe free-form shapes such as car bodies, turbine blades, ship hulls → **synthetic curves** are built from **data points / control points** by mathematical formulation.
- Two approaches:
  - **Interpolation** – curve passes exactly through the given points (Hermite cubic, cubic spline).
  - **Approximation** – curve is shaped by control points but need not pass through them (Bezier, B-spline, NURBS).
- Desirable properties: smoothness (continuity), local control, ease of modification, low degree (cubic gives best compromise), invariance under transformation.
- **Continuity**: C0 – curves just meet; C1 – same tangent direction (slope); C2 – same curvature. (G0, G1, G2 = geometric continuity).

5.1 Hermite cubic spline

- Defined by two end points P1, P2 and two end tangent vectors P1′, P2′.

$$P(u) = F1\,P1 + F2\,P2 + F3\,P1' + F4\,P2', \quad 0 \le u \le 1$$

- Blending functions: $F1 = 2u^3 - 3u^2 + 1$, $F2 = -2u^3 + 3u^2$, $F3 = u^3 - 2u^2 + u$, $F4 = u^3 - u^2$.
- Matrix form: $P(u) = [u^3\ u^2\ u\ 1]\,[M_H]\,[P1\ P2\ P1'\ P2']^T$

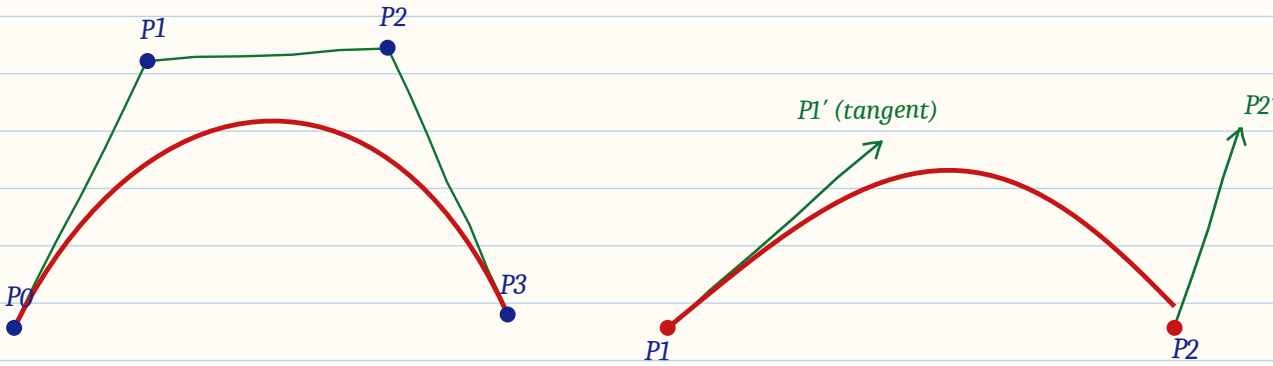
$$[M_H] = \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

- Check: $u = 0 \rightarrow P1$; $u = 1 \rightarrow P2$. Tangent magnitude controls the "fullness" of the curve; two Hermite segments joined with equal tangent → C1 continuity.
- Drawback: user must give tangents – not intuitive.

5.2 Bezier curve

- Defined by **n + 1 control points** P0 … Pn (control polygon); degree = n.

$$\mathbf{P(u) = \Sigma\ Pi \cdot B(i,n)(u)\ ,\quad B(i,n)(u) = C(n,i)\ u^i\ (1-u)^{n-i}}$$



Bezier curve (cubic) Hermite cubic

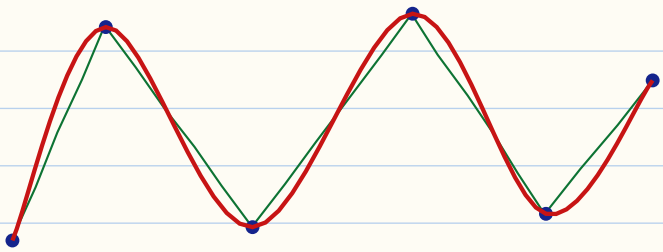
- Cubic Bezier (n = 3): $P(u) = (1-u)^3P0 + 3u(1-u)^2P1 + 3u^2(1-u)P2 + u^3P3$.
- **Properties**:
  - Curve passes through the first and last control points only (P0 and Pn).
  - Tangent at P0 is along P0P1; tangent at Pn is along Pn−1Pn.
  - Curve lies inside the **convex hull** of the control polygon; blending functions are all positive and add to 1.
  - **Variation diminishing** – no wiggles beyond the polygon; symmetric about u = 0.5.
  - Degree = number of control points − 1, so many points give a high-degree curve; **no local control** (moving one point changes the whole curve).
  - Joining segments: C1 continuity if P3 of first, P0 of second and neighbours are collinear.

5.3 B-spline curve

- Also defined by control points but made of **piecewise polynomial segments** of degree p chosen independently of the number of control points.

$$P(u) = \Sigma\, Pi \cdot N(i,k)(u), \quad i = 0 \ldots n \quad (k = order = degree + 1)$$

- Basis functions N(i,k) are found from a **knot vector** (non-decreasing set of parameter values) by the Cox–de Boor recursion formula.
- **Properties**: local control (a control point affects only k segments), degree independent of number of points, built-in continuity C(k−2), convex-hull property, curve does not pass through control points (unless the knots are repeated – clamped end conditions).
- Types: uniform (equal knot spacing), non-uniform, open (clamped) – the open B-spline passes through first and last points.



moving P3 changes only nearby segments

B-spline: local control (smooth, approximates polygon)

5.4 NURBS

- **Non-Uniform Rational B-Splines**: B-spline with a **weight** wi on every control point and a non-uniform knot vector.

$$P(u) = \Sigma\, wi\, Pi\, N(i,k)(u) \;/\; \Sigma\, wi\, N(i,k)(u)$$

- Advantages: can exactly represent **conics (circle, ellipse, parabola)** and free-form shapes in one form; weights give extra shape control (larger weight pulls the curve towards the point); invariant under projective transformation. Standard in CAD (IGES, STEP).

Comparison

Curve	Passes through points	Local control	Remark
Hermite	all end points	no	needs tangents
Bezier	first & last	no	simple, degree = n
B-spline	generally not	yes	degree independent of n
NURBS	generally not	yes	exact conics + free-form

6. Surface Modeling

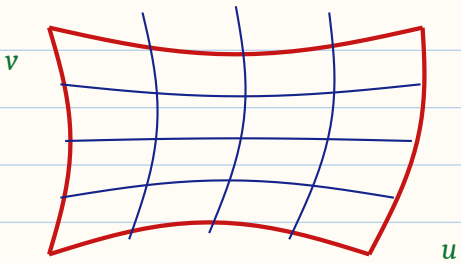
- Surface model describes the **outer skin** of an object using surface patches; it has **no volume (inside) information**.
- Used where the shape is complex or aesthetic: automobile bodies, aircraft fuselage, ship hulls, turbine blades, dies & moulds, consumer products.

Advantages and limitations

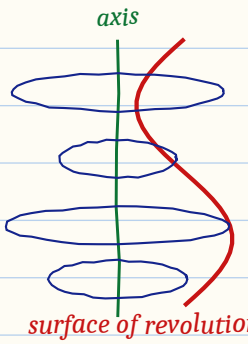
- Unambiguous shape, hidden-line / surface removal, shading, 5-axis NC tool paths, intersection of surfaces; better than wire-frame.
- Limitations: no mass properties directly; solid information missing, difficult to ensure a closed watertight body; more data and skill needed.

7. Synthetic Surfaces

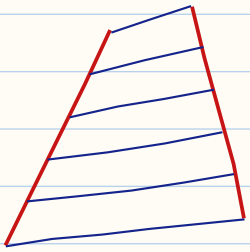
- **Analytic surfaces**: plane, cylinder, cone, sphere, torus (equations known).
- **Synthetic (free-form) surfaces**: created from points / curves – the surface analogue of curves **P(u,v)**, with two parameters u and v ($0 \le u, v \le 1$).



Surface patch P(u,v)



surface of revolution


ruled surface

7.1 Plane surface

- Defined by three non-collinear points or a point and a normal: $P(u, v) = P0 + u(P1 - P0) + v(P2 - P0)$.

7.2 Ruled (lofted) surface

- Generated by joining corresponding points of two boundary curves P(u) and Q(u) with straight lines (rulings):

$$S(u, v) = (1 - v)\, P(u) + v\, Q(u)$$

- Examples: cone, cylinder, hyperboloid; used for turbine blades.

7.3 Surface of revolution

- A plane curve (**generatrix**) is revolved about an axis through angle θ: $x = f(u)\cos\theta$, $y = f(u)\sin\theta$, $z = g(u)$.
- Examples: vases, pulleys, bottles, domes, lathe-turned parts.

7.4 Tabulated cylinder

- A plane curve is translated along a straight line (direction vector) – it is the surface of a translational sweep (extrusion).

7.5 Hermite bicubic surface

- 16 vectors: 4 corner points, 8 tangent vectors (u and v at corners) and 4 twist vectors.

$$P(u, v) = \Sigma\, \Sigma\ aij\, u^i\, v^j \quad (i, j = 0 \ldots 3)$$

- Patch can be joined with C1 continuity to neighbours.

7.6 Bezier surface

- Uses a **rectangular net** of $(m + 1) \times (n + 1)$ control points.

$$P(u,v) = \Sigma i\, \Sigma j\, Pij\, Bi,m(u)\, Bj,n(v)$$

- Properties are the extension of the Bezier curve: corner points are on the surface, lies inside the convex hull of the net, edge curves are Bezier curves.

7.7 B-spline and NURBS surfaces

- Tensor product of B-spline basis functions in u and v with knot vectors U and V → local shape control; NURBS adds weights → exact spheres, cylinders, tori.

7.8 Coons patch

- Surface built by blending four boundary curves P(u,0), P(u,1), P(0,v), P(1,v) – used to fill a closed boundary (**surface blending**).

Other surface operations

- **Fillet / blend** surfaces, **offset** surface, trimming, **sweep** and **loft** (skinning through several sections), **extension**, intersection.

8. Solid Modeling

- **Solid model** = complete, unambiguous, informationally complete representation of the volume: it knows which points are **inside, outside or on the boundary** of the object.
- Basic requirements (valid solid): **rigid** (shape independent of position), **homogeneous (3D)** – no dangling edges / faces, **finite** (bounded), closed under rigid motion and Boolean operations, **finitely describable**, boundary determinism.
- Advantages: unambiguous, mass properties (volume, centroid, inertia), interference check, section views, FEA mesh, NC verification, assembly modeling.
- Disadvantages: needs more memory and computing power; free-form shapes more difficult.

Solid representation schemes

- **Boundary representation (B-rep)**, **Constructive solid geometry (CSG)**, **Sweep representation**, **Spatial enumeration (voxels / octree)**, **Cell decomposition**, **Primitive instancing**, **Feature-based (parametric history) modeling**.

9. Fundamentals of Solids – Primitives and Boolean Operations

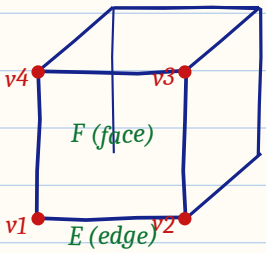
- **Primitives** (basic solids): block / box, cylinder, cone, sphere, torus, wedge, pyramid.
- **Boolean (regularised set) operations**:
  - **Union (∪)** – joins A and B into one solid.
  - **Intersection (∩)** – common volume of A and B.
  - **Difference (–)** – removes B from A (hole, slot, pocket).
- Regularisation ensures the result is again a valid solid (no dangling faces / edges).
- Other: **sweeping** (extrude, revolve, helical sweep), **blending / filleting**, **chamfering**, **shelling**, **draft**.
- **Euler's formula** for a valid simple polyhedron:

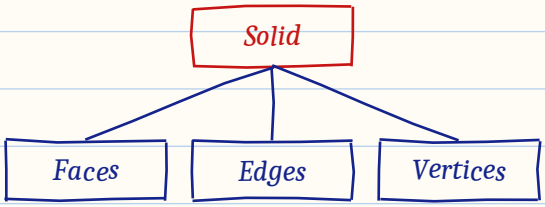
$$V - E + F = 2 \quad \text{(cube: } 8 - 12 + 6 = 2\text{)}$$

- For solids with holes (genus g): $V - E + F - R = 2(S - g)$ (R = inner loops, S = shells).

10. Boundary Representation (B-rep)

- Solid is described by its **boundary** – set of **faces** bounded by **edges** that end at **vertices**; the surface separates inside from outside.
- Data stored in two parts:
  - **Geometry** – equations of surfaces, curves and coordinates of vertices.
  - **Topology** – how faces, edges, vertices are connected (adjacency), and the orientation (outward normal) of each face.





Topology: F, E, V relations

Euler: V – E + F = 2

- Typical structure (winged-edge): Solid → Faces → Loops → Edges (with two adjacent faces and two end vertices) → Vertices.

Advantages

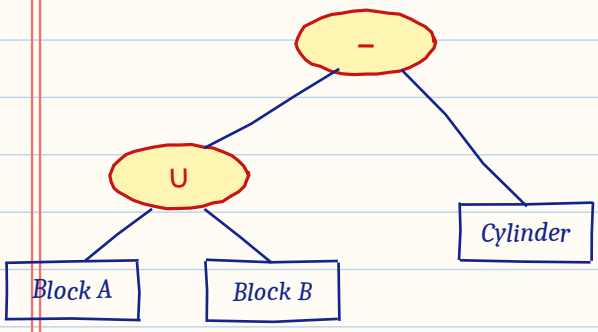
- Fast display and hidden-line removal (faces are explicit); good for free-form surfaces; easy conversion to wire-frame; direct link to surface modeling; good for NC.

Disadvantages

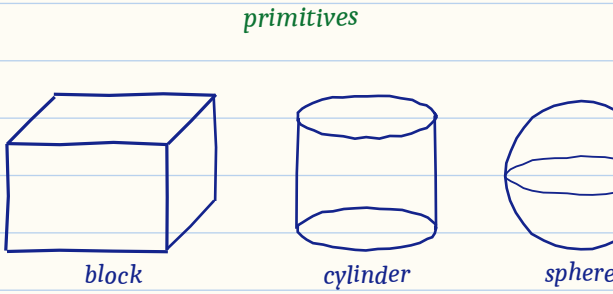
- Large storage; complex data structure; validity must be maintained (Euler operators); cannot directly show the design history; Boolean operations are complex.

11. Constructive Solid Geometry (CSG)

- Solid is built by combining **primitive solids** with **Boolean operations** and rigid motions; stored as a **binary tree**: leaves = primitives (with size & position), nodes = operations (∪, ∩, −) or transformations.



CSG tree: (A ∪ B) – Cylinder



- Evaluation of the tree gives the final object; changing a primitive's size (e.g. hole diameter) is easy (parametric).

Advantages

- Compact data, always valid solids, simple to create, easy modification, records design history, easy to compute volume & mass.

Disadvantages

- Only the tree is stored → boundary (faces / edges) must be **evaluated** for display (slow); limited to Boolean operations, difficult to create sculptured surfaces; not unique (the same solid can have different trees).

B-rep vs CSG

B-rep	CSG
Stores faces, edges, vertices	Stores primitives + Boolean tree
Large data; fast display	Compact; slow display (needs evaluation)
Easy free-form shapes	Limited to primitives
Needs validity checks	Always valid
Boundary evaluated already	Boundary must be computed

- **Hybrid modelers** (CSG + B-rep) are used in practice (CATIA, NX, SolidWorks): the CSG tree is the master and B-rep is kept for display.

12. Other Solid Schemes (brief)

- **Sweep representation**: a 2D profile moved along a path (translational, rotational, general sweep). Good for extruded / turned parts.
- **Spatial enumeration**: space divided into small cubes (voxels); **octree** divides recursively – used in medical imaging and 3D printing slicing.
- **Cell decomposition**: solid built from simple cells glued together – FEM meshes.
- **Feature-based modeling**: parametric features (hole, slot, rib, fillet) with dimensions

and constraints; changing a dimension regenerates the model.

12A. Comparison of the Three Modeling Methods

Feature	Wire-frame	Surface	Solid
Information stored	vertices & edges	edges + surface patches	faces, edges, volume, inside / outside
Ambiguity	high	low	none
Hidden line removal	not possible / manual	possible	automatic
Mass properties	no	no (approx.)	yes – volume, centre of gravity, inertia
Memory & time	least	medium	most
NC / FEA use	poor	good (surface machining)	excellent
Typical use	layouts, skeleton, mechanisms	car body, blades, dies	machine parts, assemblies

12B. Worked Examples

Example 1 – Bezier curve point

- Control points P0(0,0), P1(2,4), P2(5,4), P3(7,0); find the point at $u = 0.5$.
- Blending values ($n = 3$, $u = 0.5$): $(1-u)^3 = 0.125$, $3u(1-u)^2 = 0.375$, $3u^2(1-u) = 0.375$, $u^3 = 0.125$.
- $x = 0.125(0) + 0.375(2) + 0.375(5) + 0.125(7) = 0.75 + 1.875 + 0.875 =$ **3.5**
- $y = 0.125(0) + 0.375(4) + 0.375(4) + 0.125(0) = 1.5 + 1.5 =$ **3.0**

P(0.5) = (3.5, 3.0) – lies inside the convex hull, as the property says.

Example 2 – Hermite curve

- P1(0,0), P2(4,0), tangents P1′ = (0,4), P2′ = (0,–4). At $u = 0.5$: F1 = 0.5, F2 = 0.5, F3 = 0.125, F4 = –0.125.
- $x = 0.5(0) + 0.5(4) + 0.125(0) - 0.125(0) = 2$.
- $y = 0 + 0 + 0.125(4) - 0.125(-4) = 0.5 + 0.5 = 1$.

P(0.5) = (2, 1) → the curve bulges upward between the two end points because of the tangents.

Example 3 – Euler's formula check

- Cube: $V = 8, E = 12, F = 6 \rightarrow 8 - 12 + 6 = 2$ ✔.
- Triangular prism: $V = 6, E = 9, F = 5 \rightarrow 6 - 9 + 5 = 2$ ✔.
- Cube with a through-hole (genus 1): $V = 16, E = 24, F = 10 \rightarrow V - E + F = 2 - 2g = 0$ ✔ (so the simple formula is changed for holes).

Example 4 – Parametric equations

- Line from A(2,1) to B(6,5): $x = 2 + 4u, y = 1 + 4u, 0 \le u \le 1$.
- Circle of radius 5 at origin: $x = 5 \cos u, y = 5 \sin u, 0 \le u \le 2\pi$.
- Ellipse a, b: $x = a \cos u, y = b \sin u$. Helix: $x = r \cos u, y = r \sin u, z = pu/2\pi$.
- Parabola: $x = u, y = a\,u^2$ (non-parametric $y = a\,x^2$).

12C. Features and Parametric Modeling (modern practice)

- **Feature** = an engineering shape with meaning (hole, slot, pocket, boss, rib, fillet, chamfer, thread) – created with dimensions, not with raw primitives.
- **Parametric (history-based) modeling**: the model stores the sequence of operations (feature tree) and dimensions; changing a dimension regenerates the part.
- **Constraints**: geometric (parallel, tangent, concentric, perpendicular) and dimensional (length, angle, diameter) – make a sketch fully defined.
- **Variational vs parametric**: variational solves all constraints simultaneously (no fixed order); parametric solves in order of creation.
- **Assembly modeling**: mates / constraints between parts (coincident, concentric, distance), interference check, exploded views, bill of materials.
- **Direct modeling**: push / pull faces without the history tree – quick edits of imported geometry.
- **Data exchange**: IGES (curves and surfaces), STEP (full product data), STL (triangle mesh for 3D printing), DXF (2D drawing).

13. Quick Revision

Representation: explicit $y = f(x)$, implicit $f(x,y)=0$, parametric $P(u)$.

- Curves: Hermite (end points + tangents), Bezier (control polygon, no local control), B-spline (local control), NURBS (weights, exact conics).
- Continuity: C0 position, C1 tangent, C2 curvature.
- Surfaces: plane, ruled, revolution, tabulated cylinder, Hermite bicubic, Bezier,

B-spline, NURBS, Coons.

- Wire-frame < Surface < Solid (in information content). Wire-frame is ambiguous; solid is unambiguous.
- Solid schemes: B-rep (faces, edges, vertices; Euler V−E+F=2) and CSG (primitives + Boolean tree).

Likely exam questions

- Compare parametric and non-parametric representation of curves.
- Derive the Hermite cubic curve equation with blending functions.
- Explain Bezier curve, its properties and advantages over B-spline.
- Write short notes on wire-frame modeling: advantages and limitations.
- Explain the various synthetic surfaces with sketches.
- Differentiate between B-rep and CSG solid modeling schemes with neat figures.
- Explain Boolean operations and Euler's formula with examples.